



Leveraging service recovery strategies to reduce customer churn in an emerging market

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Abstract

Building on the properties of emerging markets, we investigate how a firm should align its service recovery strategies with different types of service failure to reduce customer churn in an emerging market. Using resource exchange theory and a multi-method approach, we show that the conventional wisdom related to service recovery needs to be reevaluated in emerging markets. Our results show that process failures lead to a higher likelihood of customer churn compared to outcome failures in emerging markets. Investigating service recovery mechanisms, we find that compensation is more effective in recovering from process failures than in recovering from outcome failures in emerging markets. Similarly, employee behavior has a stronger impact on mitigating the ill effects of process failures than those of outcome failures. The study contributes to the literature on service recovery and resource exchange theory and provides managerial insights for the effective management of customer churn due to service failures in emerging markets.

Keywords Service recovery · Emerging markets · Type of failure · Multi-methods

Introduction

Regardless of the quality of a firm's products or services, every service organization eventually faces a critical question:

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how can the effects of service failures be mitigated? Addressing this question is particularly important for firms in emerging markets that are striving for sustainable performance in the face of high service failure-related churn. Opportunities to recover from service failure are abundant, and firms can win customers' hearts by going beyond the call of duty. Consider how Club Med-Cancun (part of Club Méditerranée) recovered from a terrible service nightmare and won over a group of vacationers (Hart et al. 1989). Although the vacationers' flight took 10 hours longer than planned and ran out of food and drinks, the smart service recovery mechanisms (e.g., food at the airport, personalized greetings, live music, a sympathetic ear, a chauffeured ride, and so on) adopted by the resort helped them to better manage the service experience of the vacationers and to win the battle for market share simply by implementing a focused service-recovery strategy. It is tempting to ignore occasional complaints (e.g., billing issues for cellular service, error in a hotel room booking, or the need for more salt or cheese in a meal) and ill-tempered customers, but should managers do that? Is it possible for managers to increase firm performance (e.g., increasing profits or reducing churn) by effectively using service recovery mechanisms to handle service failures?

Firms that alienate and dissatisfy their customers will have a lot to lose, but firms that attempt to recover from service

failure will have much more to gain. For example, a recent BBC report has documented how service failures have led many to boycott companies such as Tesco, British Airways, etc.¹ Similarly, Npower in the UK lost 300,000 customers within a single year due to billing errors and poor service recovery.² By contrast, many firms such as Southwest Airlines have reaped the fruits of effectively managing service failures.³

Although multiple academic and managerial studies explore the linkages among service failures, recovery mechanisms, and customer churn (Grainer et al. 2014; Knox and van Oest 2014; Smith et al. 1999), insights are largely drawn from the context of developed market firms and are based on either theoretical understanding or behavioral/qualitative research. The literature notes that not all failures are the same, and the impact of different types of failures on customer churn can be significantly different. Accordingly, service failures are categorized as *process failures* (failures associated with the service delivery process) and *outcome failures* (failures associated with the outcome of a service) (Smith et al. 1999). Table 1 explains the notable differences between process and outcome failures.

Subsequently, scholars have proposed that service recovery strategies (*actions taken by a firm after service failures*) must match the type of failure, and they consider compensation, response speed, and employee behavior as three commonly deployed service recovery mechanisms (Gelbrich and Roschk 2011b). These mechanisms are considered fundamental to service recovery because consumers evaluate service recovery efforts in terms of the outcome of service recovery (compensation), the process of service recovery (response speed), and the interpersonal treatment (polite behavior) of consumers during the recovery process (Blodgett et al. 1997; Karande et al. 2007). Studies argue that compensation and response speed are the most effective for recovering from outcome failures, whereas employee behavior is the most effective for recovering from process failures (Smith et al. 1999).

Although such insights are important, the service recovery literature suffers from a few major limitations. First, most research in this space has been carried out in developed markets in the United States and Western Europe. It is not clear whether insights generated based on developed markets will be replicable in emerging markets, which are different from developed markets in terms of both structural properties and cultural norms (Sheth 2011; Burgess and Steenkamp 2006). Both of these factors may affect a customer's recovery expectations and how consumers exchange resources; thus, findings from developed markets may not be applicable in emerging

markets. Interestingly, the volume of service failures is greater in emerging markets. A recent study by the global consulting firm Accenture notes that approximately 88% of Indian consumers switch providers due to customer service-related issues, and the churn rate associated with service failures is much higher than the global average of 64%.⁴ Similarly, in Latin America, the top four mobile network operators have suffered revenue losses of \$30 bn dollars due to customer churn.⁵

Second, research in service recovery has not made substantial progress toward outlining a generalizable framework. Indeed, a lack of real-world data to analyze failure and recovery has led researchers to focus on laboratory experiments presenting findings that may not mimic the real world (Gelbrich and Roschk 2011b). Therefore, scholars have repeatedly called for more empirical research using real-world data that can unravel the nuances associated with service failure and recovery (Knox and van Oest 2014; Gelbrich and Roschk 2011b).

Finally, even the existing research on service failure, recovery and customer churn has been called into question by multiple scholars. Some scholars have argued that recovery strategies should be altered for different types of failure (Smith et al. 1999), while others question these relationships (Zhu et al. 2004). Because emerging markets are structurally and culturally distinct, they present us with a context in which to closely examine these alternative viewpoints in the literature.

Therefore, to address these gaps in the literature and provide actionable intelligence to managers, our study investigates three research questions: (1) How do different types (process vs. outcome) of service failures influence customer churn in emerging markets? (2) How can firms leverage recovery strategies (e.g., compensation, politeness and courtesy, and response speed) to reduce customer churn by aligning them with service failures? (3) How can service failure and recovery mechanisms be better managed in emerging markets?

We answer these research questions based on (1) *resource exchange theory* (which addresses how a consumer values and trades resources; Foa and Foa 1974) and the resource constraints of emerging market consumers (Zhu et al. 2004) and (2) the structural (Sheth 2011) and cultural (Burgess and Steenkamp 2006) properties of emerging markets.

We deploy a multi-method approach to rigorously test our hypothesis (Hamilton 2016) and mitigate concerns raised by earlier scholars about the validity of our findings. In Study 1, we conduct a qualitative study by interviewing practicing managers in emerging markets to understand the managerial

¹ <https://www.bbc.com/news/business-41320615>.

² <https://www.telegraph.co.uk/finance/personalfinance/energy-bills/11802872/Npower-loses-300000-customers-and-profits-plummet-thanks-to-billing-errors.html>.

³ <http://www.airlinerratings.com/news/449/how-southwest-handles-complaints-the-social-media-connection>.

⁴ <https://www.thehindu.com/news/cities/mumbai/Customers-head-for-the-exit-at-first-sign-of-poor-service/article10026653.ece>.

⁵ <http://www.mahindracomviva.com/mahindra-comviva-estimates-mexico-incurred-highest-revenue-loss-due-to-churn-in-latin-america.html>.

Table 1 Differences between process and outcome failure

	Outcome	Process
Nature of failure	Failure in the service outcome (Core Service)	Failure in the service process (Service Delivery Process)
Type of loss	Economic	Social
Examples	Cancellation of flights, billing errors, denial of service in a hotel because of overbooking, providing the wrong hotel room	Flight delays, long queues, rude behavior, inattentive service

perspective associated with service failure and recovery. Building on the insights from Study 1, we develop a conceptual model of service recovery rooted in the *theory of expectation* in the context of a consumer in an emerging market with its idiosyncratic structural and cultural properties. We test this model with real-world data from a telecom service provider (Study 2) and a set of behavioral experiments performed in two separate service industries (Study 3). Thus, we combine the best of both worlds (real-world data and behavioral experiments) to generate our insights. Contrary to conventional wisdom, we find that *process failures lead to a higher likelihood of customer churn than outcome failures* in emerging markets. *Compensation* is more helpful in recovering from process failures than in recovering from outcome failures. *Politeness and courtesy* positively moderate the relationship between process failure and customer churn. These results are congruent with the proposed model of behavior of an emerging market consumer.

We organize this study in the following manner. In the second section, we discuss Study 1 (qualitative study). We discuss the theoretical background, hypotheses, and conceptual framework in the third section. In the fourth section, we introduce Study 2, and in the fifth section, we describe the results. We detail Study 3 (experiments) in the sixth section. Finally, we discuss our contributions to both academia and practice and thereafter outline limitations and suggest avenues for future research.

Study 1: Qualitative study

The objectives of this qualitative study are (1) to understand current recovery strategies adopted by managers and validate that our research questions are indeed important from the implementation perspective and (2) to inform us about how firms look at service failures (e.g., service failure as one construct vs. different types of service failure) and recovery mechanisms in emerging markets. The qualitative study has helped us obtain interesting insights and understand the status of service recovery in emerging markets.

To understand managers' perspectives about service failures and recovery strategies in emerging markets, we interviewed 47 managers from four service industries. We

obtained contact information for 72 managers from three sources: (1) participants in a management development program in a premier business school in India, (2) participants in a managerial training program, and (3) industry references from one of the authors. Our selection of managers was based on the following three criteria: (1) managers must be responsible for increasing customer satisfaction in their organizations, (2) they should have at least 5 years of experience in service management in the same organization, and (3) they must oversee emerging market businesses. We ensured that we interacted with managers from different levels of management working with firms of different sizes—from small start-ups to large multinationals—to understand how managers' perspectives on service failure and recovery strategies may differ within an industry. The industries included in the study are hospitality, banking and financial services (BFSI hereafter), retailing, and telecommunications. These sectors account for approximately 35% of India's total GDP⁶⁷⁸ (Gupta and Malik 2012), making them important for our context. We contacted the managers in person during their programs and asked them to participate in the study. Fifty managers agreed to participate. In the next step, we conducted both telephonic and face-to-face interviews with 47 managers. The managers occupied different levels of management (19 senior managers, 15 mid-level managers, and 13 junior managers).

We recorded the interviews with the consent of the managers. Consistent with Holloway and Beatty (2003) and Besharov (2014), one of the authors and an independent coder independently interpreted the interviews and wrote memos reflecting their interpretations. Both the author and the independent coder individually coded the interviews, constructed empirical themes, and compared these themes both within and across transcripts. The memos contain dominant themes generated by the author and the independent coder. The author and independent coder then compared their memos and addressed any minor disagreements. The author held regular meetings with the independent coder to further reevaluate

⁶ <https://www.daedalus.co.in/financial-services-market-india/>.

⁷ <https://www.financialexpress.com/economy/tapping-the-telecom-sector-for-next-phase-of-gdp-growth/980877/>.

⁸ <http://www.businessworld.in/article/Hospitality-Industry-In-India-A-Big-Contributor-To-Economy-s-Growth-/16-05-2017-118291/>.

the themes. The validity of the codes was confirmed by presenting them to four managers (1 from each industry) (see [Web Appendix for Qualitative Research Procedure](#)).

Our interviews were guided by four major issues related to (1) managers' understanding of service failure in their organization, (2) the consequences of service failures experienced by the firm, (3) types of recovery strategies adopted by the firm, and (4) managers' strategies of categorizing different failures differently and matching recovery strategies with types of failures.

Based on the qualitative study, the following insights emerged.⁹ With respect to managers' understanding of service failures (Theme 1), we find that *managers adopt a skeptical view of their customers when working on service failure and recovery. Managers view complaints associated with service failures as problems rather than as opportunities*. The Vice President of a telecom firm noted, "we do not have a clear strategy and many customers complain no matter what. Managing failures is expensive and we do so based on our experience." Another manager from the BFSI sector stated, "Service failures are there and we try to handle them with the limited resources we have, but customers are not happy with what we do." This skepticism can have dire consequences, as managers may not be able to address service failures effectively.

Regarding the consequences of service failure (Theme 2), we find that *"customer churn" is considered a major consequence by more than 90% of managers*. Similarly, in the BFSI sector, churn and negative word-of-mouth are serious consequences of service failure. One of the managers from the BFSI sector said, "that customer was fighting about the 30-rupee fees we added for late payment. I know my team added it by mistake and we cancelled them. She switched the next month and sent a nasty email to our head." Despite the dire consequences associated with service failures, there is a lack of consensus among managers about which recovery mechanism is the most effective for combatting the ill effects of service failures.

Regarding current recovery mechanisms and their drawbacks (Theme 3), *most managers acknowledge that current recovery strategies are inadequate*. In particular, in start-ups and small organizations, managers may not have a recovery strategy—even if they do, they do not have a matrix to analyze the effectiveness of their strategies. However, most managers agreed that recovery strategies must be attuned to address the realities of emerging markets.

Regarding "aligning service recovery strategies with service failure" (Theme 4), *most managers have expressed positive views about categorizing failures and providing solutions based on types of failure*. One of the managers from the hospitality industry believes that "such a strategy can reverse our fortunes in a market that is extremely competitive in terms of price." Again, managers from the retail industry feel that it is

critical to provide solutions based on the type of failure. One of the managers responded, "If it is done properly and it gives me the actionable intelligence to win back a customer, of course it has a lot of virtue in it." There is also support from managers across industries for the need to develop a deeper understanding of such a strategy.

We then consulted with our data-providing firm for additional insights into managing service failures. The understanding of managers from the data-providing firm mimics the overall understanding of the market. These managers emphasized that they are lacking in understanding of whether service failures have any alignment with the recovery strategies that reduce customer churn. Managers across industries seek effective strategies that can help them reduce customer churn by managing service failures efficiently.

In summary, managers in emerging markets are looking for strategies to combat customer churn based on types of failure, and thus we next discuss the theoretical lens that can describe the potential link among service failure, recovery strategies, and customer churn in the emerging market context. Accordingly, we develop our hypotheses and then propose the conceptual model that we test in our study.

Theoretical background and hypothesis development

Customer expectations are central to the behavioral models of service failure and recovery strategies that have been developed in the literature (Boulding et al. 1993; Magnini et al. 2007; Zeithaml et al. 1993). Expectations are standards against which a service is judged (Oliver 1980; Oliver et al. 1997). Service failures occur when service performance falls below a customer's expectations (Hess et al. 2003). Customers use expectations to form judgments about service quality and decide whether to continue with the service provider after a service failure (Anderson et al. 1994; Gronroos 1990), which makes managing expectations a critical part of the service delivery process. Prior research shows that customers have expectations for both service performance (Hess et al. 2003) and service recovery (McCollough et al. 2000). Prior research also outlines that customers have expectations of a minimum tolerable service level (referred to as an "adequate level of expectation") (Zeithaml et al. 1993) and if service delivery falls below this level, they may exit. The adequate level of expectations is influenced by multiple factors including perceived available alternatives (Boulding et al. 1993; Zeithaml et al. 1993), culture (Kopalle et al. 2010; Mattila 2001), and *resource constraints* (which may be social, psychological or economic) that the customer faces (Bendapudi and Berry 1997).

Because these factors are integral parts of the structural (Sheth 2011) and cultural (Burgess and Steenkamp 2006) properties of emerging markets, we expect that a consumer's

⁹ Details of the comments and the themes can be provided upon request. A snapshot of the comments is presented in the [Web Appendix](#).

expectations of recovery may differ in emerging markets. Structurally, high levels of market heterogeneity in income, unbranded competition, and poor infrastructure (Sheth 2011) and culturally, embeddedness and the hierarchical nature of the society (Burgess and Steenkamp 2006), may influence customer expectations. Accordingly, recovery strategies must be attuned to the realities of emerging markets.

Types of failure and recovery mechanisms: The existing literature and theoretical paradoxes

The extant marketing literature has examined types of failures and associated recovery strategies (Gelbrich and Roschk 2011b; Smith et al. 1999; Zhu et al. 2004). Failures have been classified as *process failures* and *outcome failures* (Smith et al. 1999). Failures associated with the non-availability of services are generally classified as outcome failures, whereas delayed service, long queues, breaking commitments, rude behavior, etc., are classified as process failures (Sivakumar et al. 2014). In general, outcome failures are associated with *utilitarian exchanges*, whereas process failures involve *symbolic exchanges* (Smith et al. 1999), which implies that the losses associated with process failures are primarily social in nature and the losses associated with outcome failures are primarily economic in nature. Accordingly, the extant literature suggests that firms should rely upon different recovery mechanisms to mitigate the ill effects of service failures and meet customer recovery expectations (Roschk and Gelbrich 2014; Sivakumar et al. 2014; Smith et al. 1999; Zhu et al. 2004). Relying on the concept of the matching hypothesis (Chuang et al. 2012), the extant literature demonstrates that recovery strategies that match the type of failure lead to higher post-recovery satisfaction (Chuang et al. 2012; Smith et al. 1999).

The theoretical justification for the matching hypothesis is based on how resources are valued and categorized (Smith et al. 1999). Building on the *resource exchange theory*, studies have shown that customers place a higher value on exchanges involving similar (proximal) resources than on those involving dissimilar (distal) resources (Smith et al. 1999). The resource exchange theory proposed by Foa and Foa (1974) argued that resources can be broadly classified into two dimensions: particularism and concreteness. Particularism indicates that the worth of a resource varies with its source (who provides the resource) and concreteness specifies the tangible nature of a resource. Based on the type of resources exchanged, consumers derive either economic or socioemotional benefits (Cropanzano and Mitchell 2005). For example, a resource such as money provides more economic benefits, whereas a resource such as love provides more socioemotional benefits. In the context of service failures, scholars argue that because outcome failures are associated with tangible losses, providing economic resources as a part of recovery will be more effective. Similarly, in case of

process failures, because there are social losses, recovery strategies that provide socioemotional benefits such as an apology will be more valuable. However, some scholars have questioned whether such insights hold true under all situations. Because resource exchanges are governed by the abundance/scarcity of a resource (Brinberg and Wood 1983) and prevalent norms (Cropanzano and Mitchell 2005), we argue that one must re-evaluate how resources are exchanged under such situations. Although no research in this domain has looked into this aspect theoretically, some scholars have questioned the general applicability of the matching hypothesis for two reasons: (1) there can be a carry-over effect of a consumption activity and consumers may integrate both economic and social gains and losses from both failure and recovery (Zhu et al. 2004), and (2) the sensitivity of consumers to the outcome and process dimensions of service delivery may vary based on industry differences and situational factors (McCollough et al. 2000; Webster and Sundaram 1998). A consumer with a higher sensitivity to one type of resource may value small amounts of that resource and may substitute other resources to satisfy the overall consumption activity.

As resource exchanges are governed by the scarcity and abundance of a resource and the norms that exist in a market, one can argue that integration of *market-specific factors* is important to understand the effectiveness of various failure and recovery dimensions. Specifically, because emerging markets are different from developed markets both structurally and culturally, we argue that such differences are likely to play a vital role in shaping failure and recovery expectations and subsequent churn.

Selection of the dependent variable

We focus on customer churn as our primary variable of interest for multiple reasons. First, from a managerial point of view, our qualitative interviews indicated that most managers are more concerned about customer churn than any other consequence, such as word of mouth. In addition, there is no systematic method of understanding how to address customer churn, which prompted us to focus on customer churn rather than any other variable, such as negative word of mouth. Second, prior research has shown that customer churn is an adequate reflection of consumer behavior and affects both customer- and product market-level performance (Katsikeas et al. 2016). Reducing customer churn is also one of the central drivers of firm profitability and is at the core of most marketing expenditures (Min et al. 2016). By contrast, customer (dis)satisfaction¹⁰ is a measure of the customer mindset rather than actual behavior (Katsikeas et al. 2016). While one may argue that this mindset may reflect actual behavior, this

¹⁰ We conduct an additional robustness analysis with customer (dis)satisfaction as the dependent variable.

may not always necessarily be the case. Finally, because our context is emerging markets, most consumers may not engage in negative WOM for both structural and cultural reasons. Culturally, because of the collectivist nature of emerging markets, consumers may be inclined not to engage in negative WOM to preserve group harmony (Lam et al. 2009). Structurally, a lack of formal infrastructure such as the absence of consumer forums and a lack of Internet access may hinder a consumer's ability to engage in and the firm's ability to monitor negative WOM. Therefore, we use customer churn as our primary variable of interest.

Hypothesis development

Type of failure and customer churn Sheth (2011) has proposed five distinct dimensions unique to emerging markets: unbranded competition, a chronic shortage of resources, inadequate infrastructure, market heterogeneity, and sociopolitical governance. In addition to their structure, emerging markets are characterized by cultural embeddedness and hierarchy (Burgess and Steenkamp 2006). We argue that both structural and cultural characteristics may play a vital role in shaping consumers' expectations.

From the perspective of structural characteristics, one may argue that consumers' expectations of successful outcomes are much lower in emerging markets. Because of a chronic shortage of resources, sporadic production leads to inconsistency and non-replicability (Sheth 2011), increasing the likelihood of outcome failures. Since these failures are related to structural properties, failures are prevalent throughout the market. Thus, the alternatives to which consumers have access are likely to be of low quality in emerging markets. This issue of the low quality of available alternatives is exacerbated by the fact that there is a high level of unbranded competition (Alashban et al. 2002; Sheth 2011), with most unbranded products being of low quality (Rao et al. 1999).

In contrast to outcome failures, customer expectations related to processes are likely to be relatively higher in emerging markets, which typically are characterized by cultural dimensions of embeddedness and hierarchy (Burgess and Steenkamp 2006). In cultures characterized by embeddedness, group identity and shared goals become important (Schwartz 2006, 2011; Vaclair and Fischer 2011). Social order, respect for tradition, and restraint in expressing oneself can be critical in such cultures (Schwartz 2006). Similarly, due to the hierarchical nature of society, roles are well defined in emerging markets and there is an unequal distribution of power and resources across roles (Knafo et al. 2011; Schwartz 2006). Accordingly, social power, politeness, humility, and wealth are highly valued in emerging markets (Schwartz 2006). The extant literature has also argued that culture plays a major role in shaping customers' expectations from service encounters because of the social nature of service encounters (Czepiel

1990). Personal attention and putting customers on a pedestal are central to customer expectations and are more important than elements such as efficiency and timeliness in cultures characterized by embeddedness and hierarchy (Mattila 2001; Schmitt et al. 1994). Firms are expected to behave in accordance with cultural norms, and expectations of adequate levels of behavior are higher for the process of service delivery.

It follows from our discussion that expectations of adequate levels of recovery are lower for outcome failures than for process failures. Due to the hierarchical nature of society, customers may expect more from the firm when faced with social losses. A small glitch in processes has more consequences for the firm relative to a small shortcoming in the outcome. Thus, from a recovery expectations perspective, it may be difficult to overcome process failures. Because expectations associated with processes are much higher than expectations associated with outcomes, a firm must make an extra effort to move customers away from the loss frame. Therefore, we argue that the churn associated with process failures is higher than that associated with outcome failures.

H1: Process failures will be associated with a higher likelihood of customer churn than that associated with outcome failures in emerging markets.

Exploring the role of recovery mechanisms in emerging markets As discussed above, consumers in emerging markets may be more sensitive to process failures and have higher recovery expectations in the face of such failures. It is important to understand the nature of resources that enable successful recovery. Prior research in economics shows that the utility of a resource is determined by its scarcity (Becker 2013; Skitka and Tetlock 1992). Consequently, resource exchange theory argues that the scarcity of a resource and norms may affect the exchange relationship. However, certain norms may also make certain resources more abundant than others. Although all resources might be scarce in both emerging and developed markets, we argue that because of structural properties and cultural norms, some resources are relatively scarcer in emerging markets. Structurally, one critical difference between emerging and developed markets is reflected in terms of market heterogeneity in wealth levels (Sheth 2011). As much as 40–50% of the population in emerging markets lives below the poverty line (Burgess and Steenkamp 2006; Sheth 2011). With 15% to 70% of the population surviving on less than \$1 per day and with high unemployment rates (Burgess and Steenkamp 2006), money is one of the most scarce resources in emerging markets.

However, due to cultural embeddedness and hierarchy, resources that are more particularistic are more abundant in emerging markets. Prior research shows that in hierarchical cultures, values such as humility and politeness are cultural

norms (Smith and Reynolds 2002). Individuals are expected to behave in congruence with the cultural norms of politeness to protect group harmony (Mattila 2001; Triandis 1993). For an individual, this is an inexhaustible resource. Therefore, one could argue that such resources are abundant in emerging markets.

Because there is scarcity of universalistic resources (such as money) and an abundance of particularistic resources (such as politeness), based on resource exchange theory, we argue that service recovery strategies must be reexamined based on the structural and cultural realities of emerging markets. We discuss the differential effects of recovery mechanisms on customer churn across service failures in the next section.

Compensation While most scholars have proposed that compensation is the most effective response to outcome failures (Smith et al. 1999), we argue that compensation will have a greater effect in reducing customer churn in the case of process failures than in the case of outcome failures in emerging markets. Consumers' recovery expectations in the case of outcome failures include economic resources, whereas only social resources are included in the recovery expectations in the case of process failures (Smith et al. 1999). Due to the scarcity of economic resources compared to socioemotional resources, consumers will derive higher utility from such scarce resources. Such scarcity may affect exchange processes (Brinberg and Wood 1983) and therefore, consumers may prefer scarce resources even when the losses are socioemotional in nature. The exclusion of economic resources from the recovery expectation implies that consumers will be "pleasantly surprised" if they receive tangible compensation. Additionally, providing a scarce resource such as compensation that is not a part of the customer's service recovery expectations in the case of process failures will be useful to the consumer in the larger consumption context. Therefore, we argue that providing scarce economic resources will have higher utility for consumers in emerging markets. This effect will be more prominent in process failures than in outcome failures. However, in developed markets, consumers may not substitute social resources with tangible resources due to the relatively lower value of the latter (due to relatively less scarcity) (Gelbrich and Roschk 2011a; Smith et al. 1999). Thus, we hypothesize as follows:

H2: In emerging markets, compensation will have a greater (negative) effect on customer churn in the case of a process failure than in the case of an outcome failure.

Politeness and courtesy The importance of politeness and courtesy arises in emerging markets because of the cultural norms prevalent in emerging markets (Burgess and Steenkamp 2006). Resource exchange theory argues that norms play a vital role in defining the appropriate nature of exchange in a

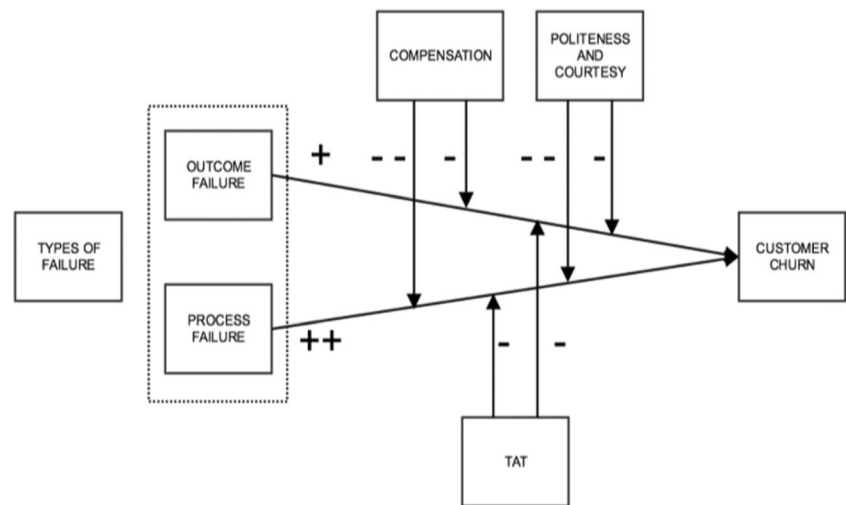
reciprocal relationship (Cropanzano and Mitchell 2005). Emerging markets are characterized by cultural norms of embeddedness and hierarchy (Burgess and Steenkamp 2006). In embedded cultures, individuals derive meaning by being a part of the larger collective group and because of the hierarchical nature of such cultures, it is vital to maintain hierarchy within a group (including power, roles and resources) (Burgess and Steenkamp 2006). The extant literature on service encounters argues that in hierarchical cultures, consumers consider service providers to be at the lower level in the hierarchy and thus, polite and courteous behavior is part of a consumer's adequate level of expectation (Mattila 1999). Drawing on earlier findings in service encounters, we argue that because politeness and courtesy are cultural norms, providing such particularistic resources may only help meet the customer's adequate level of expectations. While cultural norms make politeness and courtesy a major part of service recovery, such norms also make this resource more abundant. Coupled with hierarchy, the cultural element of embeddedness indicates that individuals should, and develop the ability to, behave according to cultural norms to maintain group harmony (Burgess and Steenkamp 2006); thus, most consumers will have an abundance of such resources. Because of the abundant nature of such resources, politeness and courtesy may have a much smaller effect than other scarce resources (Foa and Foa 1974).

From the perspective of outcome failures, such abundance is detrimental to the reduction of customer churn. Due to the tangible nature of losses that customers suffer in cases of outcome failure, providing an abundant resource such as politeness and courtesy can only be of incremental value. Again, particularistic resources are not transferable to the larger consumption activity, further diminishing the utility of this resource. By contrast, a consumer faced with a process failure expects recovery efforts that provide social value. Because politeness and courtesy are cultural norms, the provision of such resources in cases of process failure should lead to a greater impact than in cases of outcome failure. Thus, we hypothesize as follows:

H3: In emerging markets, politeness and courtesy will have a greater (negative) impact on customer churn in cases of a process failure than in cases of outcome failure.

Response speed (TAT) The role of TAT in service recovery process has been explored in detail in the literature (Gelbrich and Roschk 2011b). Two separate streams of literature present two seemingly unrelated perspectives. While some scholars argue that TAT indicates that the service provider is of high quality and failure is a rare occurrence (Wirtz et al. 2014); others argue that lower TAT signals to customers that they are important to the firm (Tyler 1989) and they have some say in the service recovery process (Tax et al. 1998).

Fig. 1 Conceptual Framework



From a resource exchange perspective, it will be a herculean task for most organizations to provide faster service recovery. Even with established processes, due to external constraints such as poor infrastructure, a firm may not be able to provide a quick recovery. This makes faster TAT a scarce resource in emerging markets. Again, from the perspective of culture, faster TAT is also a cultural norm because it indicates that customers are important to the firm, thus making it valuable in an embedded and hierarchical culture.

We hypothesize that there will not be any differential impact of TAT in cases of outcome or process failures. In both cases, consumers expect a firm to adhere to cultural norms. Again, because of the poor infrastructure in emerging markets, faster TAT is a scarce resource. Therefore, we argue that regardless of the type of failure, TAT will play a vital role, as a consumer's expectations are affected by the scarce nature of TAT. Thus, we hypothesize as follows:

H4: In emerging markets, TAT will have no differential (negative) effect on customer churn in cases of process failure compared to outcome failure.

Based on our discussion above, we present the conceptual framework in Fig. 1.

Study 2: Empirical investigation

Study 1 and the conceptual framework, which were developed based on an appropriate theoretical background, provide strong reasons to believe that the effects of different types of service failures on customer churn and the effects of recovery mechanisms may be different in emerging markets than in developed markets. However, such insights could be advanced to implementation if field data were to reveal behavior consistent with these insights. To test the conceptual framework, we need data,

and such data are not readily available. A lack of proper data may be why most studies, even in the context of developed markets, are either behavioral or conceptual. In Study 2, we conduct an empirical investigation of the conceptual framework using a proprietary dataset from a large telecom service firm.

Empirical context

Our empirical context is the telecom industry in India, which provides an excellent setting to test our hypotheses. The industry registered 7.3% compound annual growth rate over the last decade and contributes to 6.5% of the total GDP of India.¹¹ The industry is becoming increasingly concentrated in the hands of a few players in the market.¹² Customer churn is a major concern in this industry, with an average monthly churn rate of approximately 6%.¹³ Thus, the telecom industry provides a suitable context for our study. As a market, India exemplifies all the traits of an emerging market. First, a lack of infrastructure in most parts of India has presented a significant challenge for telecom companies to provide their services without interruption.¹⁴ Again, there are high levels of market heterogeneity, with the top 1% of individuals in India holding approximately 50% of the country's total wealth, and most consumers operate under tight budget constraints. India is also characterized by high hierarchy and cultural embeddedness (Schwartz 2006). Therefore, India is an appropriate context for our research.

¹¹ <https://www.financialexpress.com/economy/tapping-the-telecom-sector-for-next-phase-of-gdp-growth/980877/>.

¹² <https://www.financialexpress.com/economy/tapping-the-telecom-sector-for-next-phase-of-gdp-growth/980877/>.

¹³ <https://www.absolutdata.com/blog/can-data-analytics-stop-indias-telecom-churn-crisis/>.

¹⁴ <https://economictimes.indiatimes.com/opinion/interviews/knownskapes-rajiv-jayaraman-decodes-the-digital-blur/articleshow/64369415.cms>.

Data

We collected data from a telecom service provider that has a presence in more than 20 countries worldwide and provides services to people from all walks of life in India. The firm provided us with the data for 550 randomly selected customers¹⁵ who have experienced service failures and were provided with different recovery mechanisms. The proprietary dataset we received from the firm contained information on customer churn in the Indian market—113 out of 550 customers churned following service failure and recovery efforts. The service provider recorded the details of the service failure (i.e., a customer complaint) and the recovery mechanism initiated. Following the failure, the customer was contacted by the service provider and ratings related to TAT and employee behavior were solicited. Customers rated both of these dimensions on a scale of 1–5, where 1 indicates the worst-case scenario and 5 indicates the best-case scenario. The service provider also noted whether any tangible compensation had been provided to the customer. There were three levels of compensation: 0 (no compensation), 1 (medium compensation), and 2 (high compensation). The dataset also contains the duration of each customer's stay (length of relationship) with the firm. All of the customers were experiencing failure for the first time.¹⁶ Customer demographic-related information is hidden by the firm, as it is against firm policy to provide such information to external researchers. Note that the data we received are for customers who reported service failures.

Dependent variable

The focal firm notes whether the customer churns after the service failure and recovery mechanisms provided. Accordingly, we consider “customer churn” (yes or no) as the dependent variable. Out of the 550 service failure cases reported, 113 (20.55%) churned following service failure and recovery.

Independent variables

Based on the data provided by the firm, the authors coded the types of failures, relying on the extant literature and managerial practices in the telecom industry. The raw classification of the failures at the firm end is composed of connectivity-related issues, network coverage-related issues, billing issues, customer accounting-related issues, bill delivery-related issues, and provisioning-related issues. Based on the extant literature

¹⁵ This type of setting may sometimes create sample selection bias. To check and mitigate this potential bias, we conduct a set of behavioral experiments and find that the results are sample selection bias resistant.

¹⁶ Note that in the context of our data, customers have faced failure for the first time. We note that the exposition and modeling of repeat failure and its impact on churn is an important future research avenue given the data availability. However, in the behavioral experiments, we explicitly control for same.

Table 2 Descriptive statistics—field data

	(1)	(2)	(3)	(4)	(5)
(1) Churn	1				
(2) Compensation	0.36	1			
(3) TAT	0.28	0.16	1		
(4) Politeness and Courtesy	0.10	0.04	0.36	1	
(5) Duration of Stay	0.30	0.06	0.04	0.04	1
Mean	0.79	0.87	3.97	3.65	1.89
Std.	0.40	0.74	0.74	0.36	1.68

and communication with telecom industry experts, we classify connectivity, network coverage, billing, and accounting as outcome failures, and bill delivery and provisioning-related issues as process failures (Sivakumar et al. 2014). In most cases, there was agreement during the categorization of each failure as outcome- or process-related, but there were minor disagreements that were resolved after discussion among the authors. Once we coded outcome and process failures at our end, we confirmed our classification with a team of service experts. We matched our final classification scheme with one expert from the same firm that provided the data, and she agreed with the classification. This step removes any potential bias that may arise in categorizing service failures. Accordingly, 460 customers have faced outcome failures and 90¹⁷ have faced process failures.

Moderating and control variables

The moderating variables include perceptions of TAT and employee behavior (politeness and courtesy), which are measured on a Likert scale from 1 to 5. Compensation is classified by the firm on a low, medium, or high scale. We control for the duration of stay. We provide descriptive statistics in Table 2.

Model development

Because our dependent variable is binary, we specify a binomial logit model with a linear specification. Once a customer has faced a failure recovery scenario, s/he has the option of either churning or staying with the firm.

$$y_{iE} = 0 \text{ if customer } i \text{ chooses to churn, } 1 \text{ otherwise} \quad (1)$$

We assume that this decision is based on the utility derived from the recovery effort. We specify the deterministic component of the utility as a weighted linear combination of recovery efforts and interactions

$$y_{iE} = \alpha + \sum_k \beta_k X_{ik} + \gamma D_i + \varepsilon_i \quad (2)$$

where α is the intercept, k indexes different recovery efforts

¹⁷ Note that the percentage composition of outcome and process failures is different, which may bias the results. We test the proposed relationships with the behavioral experiments to remove any potential issues with the field data.

(politeness and courtesy, perception of time taken to resolve the complaint, compensation) for customer i , X_{ik} represents the matrix of independent and moderating variables for customer i , β are importance weights of recovery efforts, γ is the coefficient of the control variable D_i , and ε_i is a double exponentially distributed random component. The probability that a customer exits after a complaint is

$$(p_i) = \frac{\exp(\alpha + \sum \beta_k X_{ik} + \gamma D_i)}{1 + \exp(\alpha + \sum \beta_k X_{ik} + \gamma D_i)} \quad (3)$$

Results

Model estimates are presented in Table 3. The model is significant with a pseudo R^2 of 0.35. Consistent with our theoretical rationale, we find that process failures are associated with a higher likelihood of churn than outcome failures ($\beta = 11.09$, $p < 0.10$; H1 supported).

Our moderation analysis suggests that providing compensation is more effective for recovering from process failures than for recovering from outcome failures. Consumers have low expectations of compensation for process failures, but they value compensation as a recovery mechanism. The potential reason is that compensation augments a scarce resource that is useful in the broader consumption context. Medium levels of compensation are associated with a lower likelihood of churn, and this effect is higher in the case of process failure ($\beta = -13.24$, $p < 0.01$). Similarly, higher compensation is also associated with lower likelihood of churn (H2 supported). Consistent with our hypothesis, we find a stronger effect of politeness and courtesy in reducing customer churn in the case of process failure ($\beta = -3.91$, $p < 0.05$; H3 supported). However, contrary to our hypothesis, we find that TAT is more effective in case of outcome failure ($\beta = 0.97$, $p < 0.10$; H4 *n.s.*). This result may be idiosyncratic to telecom service providers and may indicate that finding a quick resolution in the case of outcome failure may have

Table 3 Parameter estimates

	Model 1 (Telecom) (Field Data)	Model 2 (Taxi Service) (Laboratory Experiment)	Model 3 (Restaurant) (Laboratory Experiment)
Independent Variables			
Type of Failure (Process Failure=0, Outcome Failure=1))	11.09* (5.90)	2.02** (0.85)	3.32*** (1.13)
Direct Effect-Moderators			
Politeness	3.64** (1.63)	0.66*** (0.11)	0.94*** (0.14)
TAT	0.008 (0.48)	0.08 (0.10)	0.28** (0.13)
High Compensation	3.33*** (0.78)	2.24*** (0.45)	3.28*** (0.50)
Medium Compensation	16.29*** (1.14)	2.06*** (0.42)	2.23*** (0.54)
Moderation Effect			
Type of Failure*High Compensation	-2.30*** (0.84)	-1.65** (0.64)	-2.66*** (0.69)
Type of Failure*Medium Compensation	-13.24*** (1.29)	-1.69*** (0.63)	-1.41** (0.70)
Type of Failure*Politeness	-3.91** (1.70)	-0.25* (0.14)	-0.49*** (0.16)
Type of Failure*TAT	0.97* (0.53)	0.29* (0.16)	0.12 (0.18)
Control Variables			
Duration of Stay	1.06*** (0.20)	-0.00 (0.01)	-0.03** (0.01)
Gender (0-Female, 1-Male)		-0.37 (0.31)	0.02 (0.29)
Propensity to Complaint		-0.01 (0.12)	0.01 (0.28)
Faced Failure Before		0.12 (0.33)	0.33 (0.34)
Frequency of usage (2) ¹		0.61 (0.41)	0.42 (0.43)
Frequency of usage (3)		0.46 (0.44)	0.09 (0.41)
Frequency of usage (4)		0.33 (0.39)	0.43 (0.43)
Frequency of usage (5)		-0.58 (0.46)	0.48 (0.60)
Age		-0.09*** (0.03)	-0.02 (0.02)
Intercept	-14.66** (5.75)	-1.30 (1.06)	-5.15*** (1.86)
AIC	389.45	441.67	404.52

***significant at 1%|**significant at 5%|*significant at 10%|standard errors in parentheses

become a priority for telecom services. We leave further assessment of the same for future research.

Turning our attention to the control variable, we find that the probability of churn decreases with an increase in duration of stay ($\beta = 1.06, p < 0.01$). This result is consistent with earlier literature, which suggests that the duration of stay reduces customer churn because consumers may place greater importance on prior satisfactory encounters than on service failures (Bolton 1998).

Endogeneity and omitted variable bias

In-depth thinking also suggests that some of the variables in our proposed model are potentially endogenous. We rely on literature, managerial practice, and our setup to understand whether the variables under consideration are endogenous.

First, the failure that a consumer faces is first-time failure. This first-time failure at the consumer level cannot be strategic from the firm's perspective as revealed by the firm; thus, this variable does not suffer from a first-order endogeneity issue.

Second, response speed (TAT) may be strategic, as a firm may have the ability to manipulate speed depending on the nature of the failure and type of the customer. To account for this endogeneity, we collect information on average response speed before 1 week from the failure incident for the i th customer. Average response speed should impact the response speed received by the customer, but it should not directly affect the customer's decision to churn. Furthermore, the average response speed and the response speed for i th customer are highly correlated (.59). However, average response speed has a low correlation with customer churn ($\rho = .11$), which shows that the instrument "average response speed" fulfills the criteria of relevance and exogeneity. Our selection of instrument was driven by both managerial insights and the extant literature.

Similarly, politeness and courtesy are potentially endogenous, as managers may decide which executive (i.e., based on past behavior, past ratings by customers, etc.) to use for each failure depending on the type of failure, which will help/hurt a customer's decision to churn. To account for possible endogeneity, we considered average peer rating (as the exclusion restriction), which may influence the quality of the executive, as reflected by the rating he/she receives.

We used the control function approach¹⁸ (Petrin and Train 2010) and estimated the first-stage equations. We find significant evidence of the impact of average response speed on the response speed ($\beta = 1.13, p < .000$) and the impact of average rating of peers on the politeness and curtesy ($\beta = .19, p < .000$). We used the residuals from the first-stage equations as the additional control variables in our proposed model. As

reflected in Table 8, we find support for our hypotheses after accounting for endogeneity.

Note that we attempted to find an instrument for compensation, as this variable is also potentially endogenous. However, we could not find any relevant data, as compensation information is very confidential to each firm. We accept this as a limitation and propose it as a future research opportunity.

We also highlight that in our empirical setup, there may be omitted variable bias, which may significantly reduce the accuracy of the estimates. Although omitted variable bias may drive endogeneity, for which we account, there are other sources of omitted variable bias. The extant literature suggests that omitted variable bias can be addressed by using robust standard errors (King and Roberts 2015). After allowing for robust standard errors in a model, if we still obtain significant estimates for our proposed relationships, we can consider that omitted variable bias is not problematic. Accordingly, we estimate our model, allowing standard error to be robust. Table 8 shows that we obtain support for our hypotheses.

Limitations of the field data

Although our field data provide useful insights into the relationship between the types of failures and customer churn and have the advantage of generalizability because they represent a sample of consumer decision making in a complex natural scenario, they come with multiple issues. Because the field data do not contain insights into underlying behavioral processes or critical variables such as severity of failure (Liao 2007; Mattila 2001; Weun et al. 2004), gender (Karande et al. 2007), prior failure (Liao 2007), frequency of use (Palmer et al. 2000), and customers who do not suffer failures, the findings may lack generalizability. Again, some consumers are more likely to complain than others (Bodey and Grace 2007). Because we only observed the customers who complain, our sample is not representative of all customers. Finally, the field data are from the mobile telecom industry, in which customers have no face-to-face contact with the service provider. Lovelock (1983) argues that in the case of such services, the outcome of the service may be more important than the processes, and our effects may be idiosyncratic to such services.

We attempt to overcome these limitations in Study 3 by conducting a set of appropriately designed experiments in which constructs of interest are measured in the context of a taxi service and a restaurant service industry. We measure recovery expectations directly and control for other factors that may act as potential confounds. Customers in the taxi service and restaurant industries not only have face-to-face interactions with the service provider (taxi drivers and the restaurant authority) but also must be physically present at the service facility (taxi and restaurant) to avail themselves of the service.

¹⁸ In a separate analysis, we attempted to account for the endogeneity using the Latent Instrument Variables approach; however, this approach did not return any fruitful results.

Study 3: Behavioral experiments

Although our field data have limitations, they maximize the realism of the study by capturing some of the aspects of the real-life story of service failure and recovery and how they are aligned to customer churn in emerging markets. To increase the precision of the results, to generalize the findings of Study 2, and overcome some of its obvious limitations, we conduct Study 3, which includes a set of scenario-based experiments (McGrath 1981). We derive the benefit of three methods that complement each other.

Insights from the taxi service industry

Research design The study employs a scenario-based experiment in which the participants are presented with a recovery scenario that they are a part of and respond accordingly. The scenario-based experiment approach is common in service failure and recovery research (Albrecht et al. 2017). The respondents were 480 graduate students from a well-known business school in India. One hundred and thirty-six respondents were female and 344 were male; the mean age was 24. These students were regular taxi users.

Subjects were initially asked to recall the app-based taxi service that they commonly use. Then, subjects were randomly assigned to 2 (outcome vs. process failure) X 3 (compensation-no compensation vs. medium compensation vs. high compensation) X 2 (TAT-faster vs. slower) X 2 (politeness-polite vs. rude) failure- recovery manipulations. While the outcome failure scenario is the unavailability of service, the process failure scenario is inattentive service (see Appendix 1 for details).

Once respondents read the failure scenario, they responded to questions related to controllability attribution (measured with a single-item scale adapted from Smith et al. 1999), severity of failure (measured with a single-item scale adapted from Mattila 2001), and recovery expectation (measured with a three-item scale adapted from Hess et al. 2003). Once the participants read the recovery scenario, they responded to a question related to the intention to churn (measured with a single item scale adapted from Singh 1990). We also obtained data on customer demographics and their assessment of how realistic the scenario was.

Pre-test and manipulation checks To check whether the failure scenarios were realistic and whether our manipulations were successful, we conducted a pre-test with 80 random subjects who did not participate in the main study. Forty-two out of 80 subjects were exposed to the outcome failure scenario and the rest were exposed to the process failure scenario. For all of our constructs, Cronbach's α is greater than 0.70, indicating a satisfactory level of reliability (Hess et al. 2003) (see Appendix 2 for details).

We also observed that our other manipulations were successful. Subjects could distinguish between polite and rude behavior conditions ($M = 5.13$, $M = 3.59$, $p < 0.05$) and faster and delayed TAT ($M = 5.64$, $M = 2.64$, $p < 0.05$). We solicited subjects' comments about the scenario once the survey was completed, and most of them felt that the scenario was believable and could happen to them in real life.

Results In the experimental setup, we controlled for severity of the failure, gender, duration of using the service, frequency of usage, age of the customer, previous failure, and propensity to complain. We again used a logistic regression to estimate the relationships of interest. One hundred and thirty-four of the 480 respondents expressed their intention to churn. Intention to churn is higher for process failure (86 out of 134 respondents) than for outcome failure (48 out of 134 respondents). We provide the descriptive statistics in Table 4.

As shown in Table 3, consistent with the results from the field data, we find support for a higher likelihood of churn for process failure than for outcome failure ($\beta = 2.02$, $p < 0.05$; H1 supported). Consistent with our conceptual arguments, we find that respondents have higher levels of recovery expectations in the case of process failure ($M = 5.43 < M = 4.61$, $p < 0.05$). Respondents feel that process failures are controllable by the firm and believe that the firm could have done something to prevent the problem ($M = 5.56 < M = 4.03$, $p < 0.05$). Similarly, process failure is also considered more severe ($M = 5.62 < M = 5.03$, $p < 0.05$). Because consumers consider process failure to be controllable by the firm, process failure is more lethal than outcome failure.

Consistent with our field data analysis, we find that compensation has a stronger negative effect on a customer's intention to churn due to process failure than due to outcome failure ($\beta = -1.69$, $p < 0.01$; $\beta = -1.65$, $p < 0.05$; H2 supported). Similarly, we find that politeness and courtesy have a stronger negative effect on a customer's intention to churn in the case of process failure than in the case of outcome failure ($\beta = -0.25$, $p < 0.10$; H3 supported). In essence, these findings indicate that compensation, a scarce resource, has a stronger

Table 4 Descriptive statistics- experiment (Taxi Service)

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Churn	1					
(2) Compensation	0.27	1				
(3) TAT	0.26	0.17	1			
(4) Politeness and Courtesy	0.37	0.09	0.30	1		
(5) Duration of Stay	-0.03	-0.03	-0.09	-0.00	1	
(6) Age	-0.08	-0.01	0.04	0.04	-0.11	1
Mean	0.72	1	4.19	4.37	19.47	24.96
Std.	0.45	0.82	1.63	1.93	11.96	3.77

effect on a customer's intention to churn in the case of process failure. By contrast, an abundant resource such as politeness and courtesy may not be adequate to move customers away from the loss frame in the case of outcome failure. These results reflect how the structural and cultural nuances of emerging markets shape customers' expectations of service recovery and behavior post recovery. These results contradict the findings from developed markets, indicating that compensation is more effective in the case of outcome failure (Smith et al. 1999). The results show that even in the experimental setting, the required alignment of service failures and recovery mechanisms is different in emerging markets than in developed markets. The results further suggest that our approach is generalizable, and not accounting for endogeneity in our field data analysis does not bias results.

Consistent with the field data and contrary to our hypothesis, we find that TAT has a stronger negative effect in the case of outcome failure than in the case of process failure ($\beta = -0.29$, $p < 0.10$; H4 *n.s.*) on customer churn (see Table 3 for details). Turning our attention to control variables, we find that customer churn decreases with increased age. All other control variables are insignificant.

Limitations of experiment 1 In Experiment 1 we could replicate our findings (from Study 2) in a laboratory setting, but it is not free from limitations. First, process failure is more severe than outcome failure. Therefore, it is entirely possible that our effects are driven by the magnitude of failure rather than the type of failure. Second, app-based taxi service, while helping us generalize our findings, has only recently been introduced in emerging markets. Therefore, it is quite possible that the majority of customers in an emerging market may not have been exposed to such services, as most consumers in such markets live either in small towns or in villages. Finally, our sample is of graduate students. Thus, it is entirely possible that our findings may not be generalizable across the larger population.

To mitigate all these concerns, we conducted a second experiment in a mid-sized city in Northeastern India. Participants were recruited through snowballing and distributing leaflets to organizations, including local businesses, schools, colleges, government offices, private businesses, etc. We used the restaurant industry as a context because after the pre-test, we realized that this is one of the most commonly used services. Finally, we controlled for the severity of failure to mitigate any alternate explanation.

Insights from the restaurant industry

Research design The research design is similar to Experiment 1, in which the participants responded to a failure recovery scenario. Four hundred and eighty respondents took part in the study, with 141 (29%) showing the intention to churn;

approximately 24% were female with a mean age of 36. There are two important characteristics worth mentioning. Consistent with the demography, the mean age is higher and the percentage of the female population is lower than the sample for the first experiment.

Subjects were asked to name a restaurant that they frequently patronize. Note that most restaurants in the area run a small-scale operation, serve similar cuisines such as breakfast and lunch, and are of standardized quality, as noted out by respondents. As in Experiment 1, subjects were randomly assigned to 2 (outcome vs. process failure) X 3 (compensation-no compensation vs. medium compensation vs. high compensation) X 2 (TAT-faster vs. slower) X 2 (politeness-polite vs. rude) failure-recovery manipulations. Consistent with Smith et al. (1999), we consider outcome failure as the unavailability of service and process failure as inattentive service (see Appendix 3 for details). Respondents were exposed to a failure recovery scenario. After respondents read the failure scenario, they responded to questions related to controllability attribution, severity of failure and recovery expectation, following which they read the recovery scenario and responded to questions related to customer churn.

Pre-test and manipulation check We conducted a pre-test among 50 randomly selected subjects with 33 males and 17 females. These subjects did not participate in the main study. Cronbach's α is greater than 0.70. We also find that subjects could distinguish between polite and rude behavior ($M = 5.08$, $M = 2.29$, $p < 0.05$) and faster vs. delayed service ($M = 5.10$, $M = 3.13$, $p < 0.05$). We solicited subjects' comments about the scenario once the survey was completed, and most subjects felt that the scenario was believable and could happen to them in real life. We controlled for the severity of failure across scenarios.

Results We provide the descriptive statistics in Table 5 and estimates in Table 3 (column 3).

Consistent with our field data, we find support for most of our hypotheses. We find that a process failure leads to a higher likelihood of churn than an outcome failure ($\beta = 3.32$, $p < 0.01$; H1 supported). We find that even after controlling for severity of failure ($M = 5.09$, $M = 4.97$, $p > 0.05$), our results hold true, which validates our assertion that in emerging markets, process failures are considered to be controllable because they contradict the cultural realities of emerging markets. We also find that compensation has a stronger effect in the case of process failure than in the case of outcome failure ($\beta = -2.66$, $p < 0.01$; $\beta = -1.41$, $p < 0.05$; H2 supported). Similarly, politeness and courtesy have a stronger effect in the case of process failure than in the case of outcome failures ($\beta = -0.49$, $p < 0.01$; H3 supported). Consistent with our hypothesis and contrary to the field data and Experiment 1, we find that TAT has no differential effect on reducing customer

Table 5 Descriptive statistics—experiment (Restaurant)

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Churn	1					
(2) Compensation	0.25	1				
(3) TAT	0.28	0.13	1			
(4) Politeness and Courtesy	0.48	0.10	0.23	1		
(5) Duration of Stay	−0.09	−0.08	−0.06	−0.03	1	
(6) Age	−0.01	−0.02	0.09	−0.01	−0.04	1
Mean	0.71	1	5.05	4.11	11.11	36.35
Std.	0.46	0.82	1.46	2.04	9.49	6.25

churn in both the case of outcome failure and the case of process failure ($\beta = 0.12$, $p > 0.1$; H4 supported). While it is surprising that the restaurant industry may show different results, this may be idiosyncratic to the industry or to the customer segment. We leave this for future research to evaluate in detail.

Turning our attention to the control variables, we find that the duration of stay has a positive effect on customer churn. Our results are consistent with the research in customer revenge-seeking behavior, which shows that duration of stay can lead to customer revenge, especially for services such as a restaurant, where there is a one-on-one connection with the customer (Grégoire et al. 2009).

Summarizing the results

In summary, through field data and two behavioral experiments across three industries, we show that in emerging markets, there is a complex interplay among failure-recovery strategies that determines customer churn. Process failure is associated with a higher likelihood of churn than outcome failure. Across all three settings, after considering multiple controls and confounds, we largely find support for our hypotheses. We find that compensation and employee behavior are more effective with process failure than with outcome failure. We perform our experiment not only with a student sample but also with a more representative customer sample to allow for a more generalized understanding. In essence, the recovery mechanisms that work in developed markets do not work in the same manner in emerging markets. Emerging market managers need to align their recovery strategies with failure types to reduce customer churn. Table 6 represents the unique contributions of the study relative to the existing literature.

Robustness check: Alternate dependent variable

Our primary dependent variable is customer churn in all cases. It is possible that our results are driven by selection of the

dependent variable. Most prior research in service recovery has used satisfaction as the primary dependent variable (Smith et al. 1999). Therefore, to mitigate any potential concerns and to be consistent with the extant literature, we also estimate our models with satisfaction (a dummy variable that takes the value of 1 for satisfaction and 0 for dissatisfaction) as a dependent variable with field data. In behavioral experiments, we asked consumers to rate their satisfaction with failure-recovery on a 7-point Likert scale. As reflected in Table 7, we find consistent results for almost all of our hypotheses. We observe one anomaly for the effect of TAT. We find that TAT has a higher effect on satisfaction in the case of outcome failure than in the case of process failure for telecom service (field data) (Table 8).

Discussion

Implications for theory

Conventional wisdom about types of failure and service recovery has long rested upon the “matching hypothesis.” Our findings, however, present an alternate perspective broadening the literature on service recovery. We find that the matching hypothesis needs to be reevaluated, keeping in mind the structural and cultural realities of emerging markets. As such, resource constraints may fundamentally change the effectiveness of multiple recovery dimensions. We find that in emerging markets, process failure is associated with a higher likelihood of churn than outcome failure. We have also shown that contrary to the earlier literature, compensation may be more effective in the case of process failure than in the case of outcome failure. Consistent with the earlier research, we find that politeness is more effective in the case of process failure than in the case of outcome failure. Our study makes three contributions to the literature.

First, we contribute to the literature on service failure by uncovering the differential effects of types of failure and subsequent recovery strategies on customer churn. The literature on service recovery has largely been agnostic to this facet (Smith et al. 1999; Sivakumar et al. 2014). We argue that because of the structural properties and cultural norms that are prevalent in emerging markets, there is a differential effect of type of failure on customer churn. To the best of our knowledge, no research has theorized on how the effect of types of failure and customer churn may be influenced by market heterogeneity and cultural norms. In doing so, we extend the literature on types of failure and customer churn by integrating market-specific contingencies.

Second, we extend the boundaries of resource exchange theory: the basic premise on which the literature on service recovery has been developed. We show that the resource exchange theory must be reevaluated by considering consumers’

Table 6 Unique contributions of the study

Studies	Type of failure considered	Markets	Theorization (Structure, culture)	Research methods	Findings
Smith et al. (1999)	Yes	Developed	Neither structure nor culture	Behavioral	Compensation and TAT are more effective in case of outcome failures; Politeness and Courtesy are more effective in case of process failure.
Zhu et al. (2004)	Yes	NA	Neither structure nor culture	Theoretical	Optimal recovery strategy depends on customer sensitivity.
Barakat et al. (2015)	No	Emerging	Culture	Behavioral	TAT is the most effective dimension of service recovery.
Gelbrich and Roschk (2011a, b)	No	Developed	Neither structure nor culture	Meta-Analysis	Compensation is most important determinant of transaction specific satisfaction, whereas employee behavior is most important determinant of cumulative satisfaction.
Our study	Yes	Emerging	Both structure and culture	Qualitative, Quantitative, and Behavioral	Process failure leads to a higher likelihood of churn compared to outcome failure in emerging markets. Compensation, politeness, and courtesy are more effective in the case of process failure than in the case of outcome failure.

resource constraints. We theorize that structural properties and cultural norms can make a resource scarcer and taking these factors into account enhances the conventional understanding related to resource exchange theory. Thus, in this work, we extend the resource exchange theory by integrating resource scarcity into the conceptual model of consumer behavior. By integrating the structural properties and cultural norms that are prevalent in emerging markets, our research challenges the assumption in resource exchange theory that the exchange of proximal resources leads to greater satisfaction (Foa and Foa 1974). We highlight that when subjected to resource constraints, even distal resources may lead to greater satisfaction. Therefore, for scarce resources, consumers may trade resources that are dissimilar and still derive greater satisfaction because of the scarcity associated with the resource. Thus, we broaden the resource exchange theory by relaxing this central assumption.

Finally, our research contributes to the literature on service recovery. Most of the existing research has relied on the cultural realities of a market (Mattila and Patterson 2004; Patterson et al. 2006) and subsequently made predictions that overlook broader structural realities. For example, while making a prediction about which recovery strategy will be more effective in emerging markets, Barakat et al. (2015) show that TAT is more effective than any other recovery mechanism. However, this research neither distinguishes between types of failure nor incorporates the structural realities of emerging markets. Incorporation of these nuances may help enhance the theory of service failure. We outline that simply considering cultural realities may lead to erroneous conclusions about the effectiveness of various recovery mechanisms. This also helps us validate the theoretical work carried out by Zhu et al.

(2004), which argues that a matching hypothesis may be altered based on the sensitivity of consumers. Our work highlights that such sensitivities can be observed in emerging markets, thus challenging the existing findings of service failure and recovery.

Implications for practice

Our research yields multiple managerial contributions. Our managerial interviews reveal that most managers in emerging markets are largely in firefighting mode in regard to service failure. Managers not only struggle with customer churn but also seek actionable intelligence. Based on our findings, managers in emerging markets may decide to form comprehensive guidelines to deal with service failures. Managers must understand that all failures are not the same, and process failure may lead to a higher likelihood of churn than outcome failure. However, recent articles in the popular press have indicated that firms are spending less in human capital development,¹⁹ which is a major reason for process failures. According to a Deloitte Insights (2013) report, 60% of companies have not invested in filling skill gaps. Based on our research, we urge organizations to make a consistent commitment to reducing these talent gaps so that process failures can be minimized.

Our second recommendation for organizations is to create champions among employees who can effectively manage either process or outcome failures rather than handling both types of failure. Most organizations in emerging markets have adopted interactive voice response (IVR) systems. However,

¹⁹ <https://www2.deloitte.com/insights/us/en/focus/business-trends/2013/emerging-market-talent-strategies.html>.

Table 7 Robustness check—parameter estimates (DV: Satisfaction)

	Model 1 (Telecom) (Field data)	Model 2 (Taxi service) (Laboratory experiment)	Model 3 (Restaurant) (Laboratory Experiment)
Independent variables			
Type of Failure (Process Failure = 0, Outcome Failure = 1))	11.25** (5.17)	4.32*** (0.48)	2.97*** (0.49)
Direct effect-moderators			
Politeness	4.06*** (1.38)	0.50*** (0.06)	0.44*** (0.05)
TAT	−0.06 (0.36)	0.21** (0.08)	0.18** (0.06)
High Compensation	1.94*** (1.08)	0.91*** (0.30)	1.39*** (0.21)
Medium Compensation	14.51*** (0.59)	1.59** (0.28)	1.16*** (0.24)
Moderation effect			
Type of Failure*High Compensation	−1.05* (1.16)	−0.75* (0.43)	−1.21*** (0.34)
Type of Failure*Medium Compensation	−12.31*** (0.64)	−1.86*** (0.42)	−1.71** (0.32)
Type of Failure*Politeness	−4.02** (1.44)	−0.39*** (0.09)	−0.27*** (0.07)
Type of Failure*TAT	0.98** (0.41)	−0.15 (0.11)	−0.12 (0.09)
Control variables			
Duration of Stay	0.46** (0.11)	−0.01* (0.01)	0.01 (0.01)
Gender (0-Female, 1-Male)		−0.30* (0.19)	−0.08 (0.14)
Propensity to Complain		−0.16* (0.09)	−0.06 (0.17)
Faced Failure Before		0.14 (0.26)	0.05 (0.16)
Frequency of usage (2)		0.33 (0.30)	0.09 (0.22)
Frequency of usage (3)		−0.08 (0.31)	−0.16 (0.21)
Frequency of usage (4)		0.71** (0.29)	0.17 (0.21)
Frequency of usage (5)		0.35 (0.34)	0.40 (0.25)
Age		0.03 (0.03)	0.00 (0.01)
Intercept	−15.13** (5.00)	−0.07 (0.76)	0.40 (0.91)
AIC	480.53	1952.13	1768.55

For field data, satisfaction is coded as 0 (dissatisfied) and 1 (Satisfied). For experiments, it is measured in a Likert scale of 1 (extremely dissatisfied) to 7 (extremely satisfied)

The “frequency of usage” question allows consumers to choose how frequently they avail themselves of the service. Frequency of usage (1) (base level) = More than once a week, Frequency of usage (2) = Once a week, Frequency of usage (3) = Once every 15 days, Frequency of usage (4) = Once a month, Frequency of usage (5) = Less than once a month.

***significant at 1%|**significant at 5%|*significant at 10%|standard errors in parentheses

in most cases, such IVRs are organized based on functions (such as loss of a credit card, queries about billing-related issues, etc.). We urge managers to create separate IVRs for outcome and process failures. Employees can also be trained to handle either outcome or process failures, which will help a firm to efficiently handle service failures and reduce customer churn. Again, because employees need to be trained in one dimension (either process or outcome), a firm’s cost of training employees will decrease. Similarly, with employees improving their ability to handle one type of complaint, employee satisfaction will increase. Additionally, such a focus may reduce incidents of double deviations (failure during recovery) which is a major cause of concern for many firms (Joireman et al. 2013).

Again, regardless of failure type, a firm must handle customers in a polite and respectful manner. In a culturally embedded and hierarchical emerging market, such behavior may go a long way toward managing complaints effectively. Our research shows that not dealing with customers politely can lead to a higher likelihood of churn across failure types. However, our managerial interviews paint a grim picture in regard to handling complaints in a polite and respectful manner. A senior manager in the hospitality industry said, “We try to be polite but we don’t go beyond a certain point.” Similar concerns have also been raised by managers in the banking industry. One manager noted, “I’m not sure if the branch staff are always going to be polite.” While politeness is a “must” in any service encounter, firms cannot provide the same to customers. Many scholars in psychology and organizational behavior have shown that organizations’ HR practices may help inculcate behavior such as politeness among employees (Kuvaas 2008). Along with favorable HR policies, organizations may rely upon increases in surveillance in the context of service recovery. Prior research has shown that effective surveillance significantly reduces deviant behavior (e.g., impolite behavior) (Harris and Ogbonna 2002). Therefore, such strategies may help a firm ensure that its employees perform their job roles in a polite and respectful manner.

Although compensation as a mechanism to recover from service failure is considered opportunistic, this research warns that especially in emerging markets, it has significant value to recover from process failures. Most managers may not fully appreciate that providing compensation can greatly reduce customer churn. As one manager in the retail industry noted, “these things are very subjective and again, it’s a judgment call by our customer satisfaction leadership team, but on a day-to-day basis, we do not look at compensating. If there is something wrong from our side, we try to apologize, to accelerate shipping and to mitigate the issues, but we do not offer compensation or coupons as a way to justify their complaint or make them happy.” These comments reflect a manager’s apprehension about using compensation as an effective recovery mechanism. The empirical evidence provided in our study

Table 8 Additional robustness analyses

	Model 4: Accounting for Endogeneity	Model 5: Model with Robust Standard Error
Independent variables		
Type of Failure (Process Failure = 0, Outcome Failure = 1))	10.86** (4.97)	11.09* (5.90)
Direct Effect-moderators		
Politeness	−1.06(2.5)	3.63** (1.63)
TAT	.86(.63)	0.008(.476)
High Compensation	3.13*** (.79)	3.32*** (.78)
Medium Compensation	16.04(1660.26)	16.23*** (1.13)
Moderation effect		
Type of Failure*High Compensation	−2.08*** (.85)	−2.30*** (.83)
Type of Failure*Medium Compensation	−13.06(1660.261)	−13.23*** (1.28)
Type of Failure*Politeness	−3.84*** (1.47)	−3.91** (1.69)
Type of Failure*TAT	.94* (.56)	.97* (.53)
Control Variables		
Duration of Stay	1.07*** (.17)	1.053*** (.19)
Endogeneity		
Politeness	4.48** (2.13)	
TAT	−1.06** (.45)	
Intercept	−.80(8.26)	−14.65*** (5.75)
AIC	386.85	389.45

***significant at 1%|**significant at 5%|*significant at 10%|standard errors in parentheses

may alleviate managerial concerns about compensation as an effective recovery mechanism and subsequently reduce customer churn.²⁰

Limitations and future research directions

While our study is a rigorous attempt to address the issue of service recovery in emerging markets, it has certain limitations. First, we have found inconclusive evidence about the role played by TAT in emerging markets. Future research should attempt to find conclusive evidence for the effect of TAT. There are multiple reasons that one may hypothesize why TAT may show a differential effect. First, there may be industry-specific heterogeneity. Although telecom consumers develop an arm's-length relationship, taxi service relationships are based on short interactions with the service provider (taxi driver). By contrast, the restaurant service that we selected for our experiments is based on long-term relationships. Therefore, TAT is important in mitigating the ill effects of outcome and process failure. However, a lack of interpersonal relationships in both the taxi service and the telecom service implies that outcomes are more important than processes

(Lovelock 1983) for such services. By contrast, restaurants rely highly on developing long-term interpersonal relationships with customers. Thus, it is possible that TAT will have a stronger effect on reducing outcome failure than on reducing process failure. Second, our results may also be driven by the interaction between the customer's time orientation and the type of failure. Consumers in most emerging markets are driven by a long-term orientation and thus may lower their expectations, especially for outcomes (Kopalle et al. 2010). Because outcomes are critical in case of taxi services and telecom, lower expectations imply that faster TAT may be more effective in case of outcome failure. We urge future researchers to focus on this aspect in greater detail.

Second, our data come from only one emerging market: India. While we do believe that our findings will be generalizable across other emerging markets, retesting in other emerging markets may provide additional insights. Third, future research may consider repeat failures and generate insights into how different mechanisms work. Fourth, although we do not consider interactions among moderators, capturing such interactions may bring new insights. Our model is compensatory in nature. However, it is not obvious that this assumption will necessarily hold true. Especially in emerging markets, in which consumers are more sensitive to some recovery dimensions than to others, future research may reconsider such fundamental assumptions. Fifth, apart from conventionally employed recovery mechanisms, future research

²⁰ We have presented our results to one of the firms in the hospitality industry, which has expressed a willingness to implement our findings. We are still in negotiation with that firm about a data-sharing agreement.

may also consider other recovery mechanisms that may be idiosyncratic to emerging markets. For example, in our interviews some respondents argue that higher management interventions are a critical component in a hierarchical culture that may negatively affect customer churn. Again, adequacy of information and communication may also be highly effective in reducing customer churn from service failures, as such communication would indicate that the firm values the relationship with its customers (Liao 2007), which can be vital in emerging markets because of their structural and cultural properties. Finally, literature has recently started to focus on proactive recovery strategies such as inoculation (sensitizing customers before service failures) (Mikolon et al. 2015). Such strategies may be vital in emerging markets because of structural properties with frequent outcome failures. Future research can explore this aspect in detail.

Struggling to manage customer churn, managers in emerging markets are facing a growing challenge to mitigate the risks caused by service failures. Without academic research to rely upon, managers may not be able to mitigate the ill effects of service failures. In this study, we show that a firm must understand market-specific differences to address issues related to service recovery. Firms must also be cognizant of the differential effect of types of failure on customer churn. A firm may wish to design recovery strategies keeping in mind the idiosyncratic characteristics of emerging markets to address the problem of service failure.

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Appendix 1: Taxi Service Failure-Recover Scenario

Please recall the Taxi Service provider you generally use.

Failure Scenarios

Outcome Failure: Unavailable Service You have to travel to another city for some important personal work. You have decided to book a taxi with the same service provider. You called the taxi service provider and booked an air-conditioned (AC) taxi with free Wi-Fi service to drop you at the airport. You have booked the taxi for 10 am on the day of your journey. Finally, the day of your journey arrives. The taxi arrives on time. However, as you board the taxi you have realized that it is not the

type of taxi you booked. The taxi has neither air conditioning nor Wi-Fi service, which was promised while booking. Finally, your journey starts. Upon reaching the airport, you have decided to complain to the service provider.

Process Failure: Inattentive Service You have to travel to another city for important personal work. You have decided to book a taxi with the same service provider. You called the taxi service and booked an air-conditioned (AC) taxi with free Wi-Fi service to drop you at the airport. You have booked the taxi for 10 am on the day of your journey. Finally, the day of your journey arrives. However, the taxi arrives 15 min late. As you board the taxi, you have realized that the driver is in the middle of a telephone call. The driver finishes his call after 5 min. Finally, your journey starts. However, you have realized that the AC is not switched on. You asked the driver to switch on the AC. However, he does not pay attention to your request. After a second request, he finally switches on the AC. Upon reaching the airport, you have decided to complain to the service provider.

A typical recovery profile is described below You have called customer care and begun to narrate the entire incident. Customer service allows you to complete what you want to say. The executive replied, “We are sorry to hear about your experience. We are committed to providing excellent customer service to our customers. Let me check with the concerned department. Could you please give me some time? I will get back to you as soon as possible”. After 10 min, you have received a call from a customer service executive. The executive informed you, “We are extremely disappointed to know that your experience with us on the last trip was not satisfactory. Based on your complaint, we have investigated the problem. We accept our mistake and will try our best to avoid such mistakes in the future. We have decided to provide a 100% refund for this trip, which will be credited to your account. We hope you will give us an opportunity to serve you again”.

We changed the recovery profiles based on our intended manipulation. For compensation, customers read, “We have decided to provide a 100% refund for this trip, which will be credited to your account” (high compensation) or “We have decided to provide a 50% refund for this trip, which will be credited in your account” (medium compensation). In a no-compensation scenario, the statement was missing. Again, for TAT customers read “After 10 minutes, you have received a call from customer service executive” (faster resolution vs. “After 3 hours, you have received a call from customer service executive” (delayed resolution). Politeness and courtesy have been manipulated by many statements in

the recovery profile such as “The customer service executive interrupts you when you try to complete what you want to say” (rude behavior) vs. “The customer service allows you to complete what you want to say” (polite behavior). Similarly, the customer reads that the executive replied, “We are committed to providing excellent customer service to our customers. Let me check with the concerned department. Give me some time and I will get back to you as soon as possible” (rude behavior) vs. the executive replied, “We are sorry to hear about your experience. We are committed to providing excellent customer service to our customers. Let me check with the concerned department. Could you please give me some time? I will get back to you as soon as possible” (Polite behavior).

Appendix 2: Measures

Model Constructs

Propensity to Complain (adapted from Bodey and Grace (2007)) (1-Strongly Agree, 7-Strongly Disagree); Coefficient alpha = .85.

PC1: If there is a service failure, I will complain to the company.

PC2: If I am dissatisfied with the things I buy, I will complain about them to the shop (or other suppliers) that sold them to me.

PC3: I do not hesitate to complain if I think it is warranted to do so.

PC4: Based on my past purchasing experiences, I am likely to complain in the event of dissatisfaction or service failure.

PC5: I am inclined to complain to the service provider if I am unhappy with a service.

PC6: I am usually reluctant to complain about service regardless of how bad it is (R).

PC7: I am less likely than most people to complain about unsatisfactory service (R).

Realism of the Scenario (adapted from Liao (2007)) (1-Not at all, 7- Completely); Coefficient alpha = .78.

RS1: The extent to which this particular scenario “sounds realistic”. (R).

RS2: The extent to which this particular scenario “could happen in real life”. (R).

Turn Around Time (adapted from Tax et al. 1998). 7-point scale anchored at the endpoints (1- Strongly Disagree, 7- Strongly Agree); Coefficient alpha = .75.

TAT1: The taxi service provider was quick to resolve my problem.

TAT2: The length of time taken to resolve my problem was longer than necessary. (R).

Service Recovery Expectations (adapted from Hess et al. (2003)) (1- Strongly Disagree, 7- Strongly Agree); Coefficient alpha = .80.

SRE1: I expect the taxi service provider to do everything in its power to solve the problem.

SRE2: I do not expect the taxi service provider to exert much effort to solve the problem. (R).

SRE3: I expect the taxi service provider to try to make up for (providing the wrong taxi/inattentive service by the driver).

Controllability Attribution (adapted from Smith et al. (1999)) (1- Strongly Disagree, 7- Strongly Agree).

Do you think the service provider could have prevented the problem?

Severity of Failure (adapted from Mattila 2001 Basso and Pizzutti (2016)) (1-Unimportant, 7-important).

Based on your experience, how important do you feel the service failure was?

Politeness and Courtesy (adapted from Liao (2007)) (1- Strongly Disagree, 7- Strongly Agree).

The customer service representative was courteous to me.

Likelihood of Churn (adapted from (Singh 1990)) (1- Strongly Disagree, 7-Strongly Agree).

Based on my experience with service failure and recovery, I am going to end my relationship with the firm.

Appendix 3: Restaurant Failure-Recovery Scenario

Please recall a restaurant that you have visited recently.

Failure Scenario

Outcome Failure You go to the same restaurant for lunch. You are seated at your table and the waiter comes to take the order. You select your items from the menu and place your order. The waiter informs you that the restaurant is out of the items you have selected. You have decided to complain to the manager.

Process Failure You go to the same restaurant for lunch. You are seated at your table, but the waiter is busy talking to other waiters. Finally, after almost shouting at the top of your voice, the waiter takes the order. However, the waiter delays inordinately in delivering your food and after two reminders, finally serves you. You have decided to complain to the manager.

Service Recovery Scenario You walk to the manager and start narrating the incident. The manager allows you to complete what you want to say. The manager replied, “We are sorry to hear about your experience. We are committed to providing excellent customer service to our customers. Let me check what

I can do for you. I will get back to you as soon as possible”. After 2 min, the manager gets back to you. The manager informed you, “We are extremely disappointed to know that your experience with us in today was not satisfactory. Based on your complaint, we have investigated the problem and we have decided to provide a 100% refund for today’s lunch. We hope you will give us an opportunity to serve you again”.

We change the recovery profiles based on our intended manipulation. For compensation, customers read “we have decided to provide a 100% refund for today’s lunch” (high compensation) or “we have decided to provide a 50% refund for today’s lunch” (medium compensation). In no-compensation scenario, the statement was missing. Again, for TAT customers read “After 2 minutes, the manager gets back to you” (faster resolution) vs. “After 15 minutes, the manager gets back to you” (slower resolution). Politeness and courtesy have been manipulated by many statements in the recovery profile such as “The manager interrupts you when you try to complete what you want to say” (rude behavior) vs. “The manager allows you to complete what you want to say” (polite behavior). Similarly, the manager replied, “We are committed to providing excellent customer service to our customers. Let me check what I can do for. I will get back to you as soon as possible” (rude behavior) vs. the manager replied, “We are sorry to hear about your experience. We are committed to providing excellent customer service to our customers. Let me check what I can do for you. I will get back to you as soon as possible” (polite behavior).

Measures: All measures are similar to Experiment 1.

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References

- Alashban, A. A., Hayes, L. A., Zinkhan, G. M., & Balazs, A. L. (2002). International brand-name standardization/adaptation: Antecedents and consequences. *Journal of International Marketing*, 10(3), 22–48.
- Albrecht, A. K., Walsh, G., & Beatty, S. E. (2017). Perceptions of group versus individual service failures and their effects on customer outcomes: The role of attributions and customer entitlement. *Journal of Service Research*, 20(2), 188–203.
- Anderson, E. W., Fornell, C., & Lehmann, D. R. (1994). Customer satisfaction, market share, and profitability: Findings from Sweden. *Journal of Marketing*, 58(3), 53–66.
- Barakat, L. L., Ramsey, J. R., Lorenz, M. P., & Gosling, M. (2015). Severe service failure recovery revisited: Evidence of its determinants in an emerging market context. *International Journal of Research in Marketing*, 32(1), 113–116.
- Basso, K., & Pizzutti, C. (2016). Trust recovery following a double deviation. *Journal of Service Research*, 19(2), 209–223.
- Becker, G. S. (2013). *The economic approach to human behavior*. Chicago: University of Chicago Press.
- Bendapudi, N., & Berry, L. L. (1997). Customers’ motivations for maintaining relationships with service providers. *Journal of Retailing*, 73(1), 15–37.
- Besharov, M. L. (2014). The relational ecology of identification: How organizational identification emerges when individuals hold divergent values. *Academy of Management Journal*, 57(5), 1485–1512.
- Blodgett, J. G., Hill, D. J., & Tax, S. S. (1997). The effects of distributive, procedural, and interactional justice on postcomplaint behavior. *Journal of Retailing*, 73(2), 185–210.
- Bodey, K., & Grace, D. (2007). Contrasting “complainers” with “non-complainers” on attitude toward complaining, propensity to complain, and key personality characteristics: A nomological look. *Psychology & Marketing*, 24(7), 579–594.
- Bolton, R. N. (1998). A dynamic model of the duration of the customer’s relationship with a continuous service provider: The role of satisfaction. *Marketing Science*, 17(1), 45–65.
- Boulding, W., Kalra, A., Staelin, R., & Zeithaml, V. A. (1993). A dynamic process model of service quality: From expectations to behavioral intentions. *Journal of Marketing Research*, 30(1), 7–27.
- Brinberg, D., & Wood, R. (1983). A resource exchange theory analysis of consumer behavior. *Journal of Consumer Research*, 10(3), 330–338.
- Burgess, S. M., & Steenkamp, J.-B. E. M. (2006). Marketing renaissance: How research in emerging markets advances marketing science and practice. *International Journal of Research in Marketing*, 23(4), 337–356.
- Chuang, S.-C., Cheng, Y.-H., Chang, C.-J., & Yang, S.-W. (2012). The effect of service failure types and service recovery on customer satisfaction: A mental accounting perspective. *Service Industries Journal*, 32(2), 257–271.
- Cropanzano, R., & Mitchell, M. S. (2005). Social exchange theory: An interdisciplinary review. *Journal of Management*, 31(6), 874–900.
- Czepiel, J. A. (1990). Service encounters and service relationships: Implications for research. *Journal of Business Research*, 20(1), 13–21.
- Foa, Uriel G, and Edna B Foa. (1974). *Societal structures of the mind*: Charles C Thomas.
- Gelbrich, K., & Roschk, H. (2011a). Do complainants appreciate overcompensation? A meta-analysis on the effect of simple compensation vs. overcompensation on post-complaint satisfaction. *Marketing Letters*, 22(1), 31–47.
- Gelbrich, K., & Roschk, H. (2011b). A meta-analysis of organizational complaint handling and customer responses. *Journal of Service Research*, 14(1), 24–43.
- Grainer, M., Noble, C. H., Bitner, M. J., & Broetzmann, S. M. (2014). What unhappy customers want. *MIT Sloan Management Review*, 55(3), 31–35.
- Grégoire, Y., Tripp, T. M., & Legoux, R. (2009). When customer love turns into lasting hate: The effects of relationship strength and time on customer revenge and avoidance. *Journal of Marketing*, 73(6), 18–32.
- Gronroos, C. (1990). Relationship approach to marketing in service contexts: The marketing and organizational behavior interface. *Journal of Business Research*, 20(1), 3–11.
- Gupta, R., & Malik, P. (2012). FDI in Indian retail Sector: Analysis of competition in Agri-food sector. Internship Project Report, Competition Commission of India.
- Hamilton, R. (2016). Consumer-based strategy: Using multiple methods to generate consumer insights that inform strategy. *Journal of Academy of Marketing Science*, 44(3), 281–285.
- Harris, L. C., & Ogbonna, E. (2002). Exploring service sabotage: The antecedents, types and consequences of frontline, deviant, antiservice behaviors. *Journal of Service Research*, 4(3), 163–183.
- Hart, C. W., Heskett, J. L., & Earl Sasser, W., Jr. (1989). The profitable art of service recovery. *Harvard Business Review*, 68(4), 148–156.
- Hess, R. L., Ganesan, S., & Klein, N. M. (2003). Service failure and recovery: The impact of relationship factors on customer satisfaction. *Journal of the Academy of Marketing Science*, 31(2), 127–145.

- Holloway, B. B., & Beatty, S. E. (2003). Service failure in online retailing: A recovery opportunity. *Journal of Service Research*, 6(1), 92–105.
- Joireman, J., Grégoire, Y., Devezer, B., & Tripp, T. M. (2013). When do customers offer firms a “second chance” following a double deviation? The impact of inferred firm motives on customer revenge and reconciliation. *Journal of Retailing*, 89(3), 315–337.
- Karande, K., Magnini, V. P., & Tam, L. (2007). Recovery voice and satisfaction after service failure an experimental investigation of mediating and moderating factors. *Journal of Service Research*, 10(2), 187–203.
- Katsikeas, C. S., Morgan, N. A., Leonidou, L. C., & Hult, G. T. M. (2016). Assessing performance outcomes in marketing. *Journal of Marketing*, 80(2), 1–20.
- King, G., & Roberts, M. E. (2015). How robust standard errors expose methodological problems they do not fix, and what to do about it. *Political Analysis*, 23(2), 159–179.
- Knafo, A., Roccas, S., & Sagiv, L. (2011). *The value of values in cross-cultural research: A special issue in honor of Shalom Schwartz*. Los Angeles: Sage Publications Sage.
- Knox, G., & van Oest, R. (2014). Customer complaints and recovery effectiveness: A customer base approach. *Journal of Marketing*, 78(5), 42–57.
- Kopalle, P. K., Lehmann, D. R., & Farley, J. U. (2010). Consumer expectations and culture: The effect of belief in karma in India. *Journal of Consumer Research*, 37(2), 251–263.
- Kuvaas, B. (2008). An exploration of how the employee–organization relationship affects the linkage between perception of developmental human resource practices and employee outcomes. *Journal of Management Studies*, 45(1), 1–25.
- Lam, D., Lee, A., & Mizerski, R. (2009). The effects of cultural values in word-of-mouth communication. *Journal of International Marketing*, 17(3), 55–70.
- Liao, H. (2007). Do it right this time: The role of employee service recovery performance in customer-perceived justice and customer loyalty after service failures. *Journal of Applied Psychology*, 92(2), 475–488.
- Lovelock, C. H. (1983). Classifying services to gain strategic marketing insights. *Journal of Marketing*, 47(3), 9–20.
- Magnini, V. P., Ford, J. B., Markowski, E. P., & Jr, E. D. H. (2007). The service recovery paradox: Justifiable theory or smoldering myth? *Journal of Services Marketing*, 21(3), 213–225.
- Mattila, A. S. (1999). The role of culture in the service evaluation process. *Journal of Service Research*, 1(3), 250–261.
- Mattila, A. S. (2001). The effectiveness of service recovery in a multi-industry setting. *Journal of Services Marketing*, 15(7), 583–596.
- Mattila, A. S., & Patterson, P. G. (2004). Service recovery and fairness perceptions in collectivist and individualist contexts. *Journal of Service Research*, 6(4), 336–346.
- McCollough, M. A., Berry, L. L., & Yadav, M. S. (2000). An empirical investigation of customer satisfaction after service failure and recovery. *Journal of Service Research*, 3(2), 121–137.
- McGrath, J. E. (1981). Dilemmas: The study of research choices and dilemmas. *American Behavioral Scientist*, 25(2), 179–210.
- Mikolon, S., Quaiser, B., & Wieseke, J. (2015). Don’t try harder: Using customer inoculation to build resistance against service failures. *Journal of the Academy of Marketing Science*, 43(4), 512–527.
- Min, S., Zhang, X., Kim, N., & Srivastava, R. K. (2016). Customer acquisition and retention spending: An analytical model and empirical investigation in wireless telecommunications markets. *Journal of Marketing Research*, 53(5), 728–744.
- Oliver, R. L. (1980). A cognitive model of the antecedents and consequences of satisfaction decisions. *Journal of Marketing Research*, 17(4), 460–469.
- Oliver, R. L., Rust, R. T., & Varki, S. (1997). Customer delight: Foundations, findings, and managerial insight. *Journal of Retailing*, 73(3), 311–336.
- Palmer, A., Beggs, R., & Keown-McMullan, C. (2000). Equity and repurchase intention following service failure. *Journal of Services Marketing*, 14(6), 513–528.
- Patterson, P. G., Cowley, E., & Prasongsukarn, K. (2006). Service failure recovery: The moderating impact of individual-level cultural value orientation on perceptions of justice. *International Journal of Research in Marketing*, 23(3), 263–277.
- Petrin, A., & Train, K. (2010). A control function approach to endogeneity in consumer choice models. *Journal of Marketing Research*, 47(1), 3–13.
- Rao, A. R., Qu, L., & Ruekert, R. W. (1999). Signaling unobservable product quality through a brand ally. *Journal of Marketing Research*, 36(2), 258–268.
- Roschk, H., & Gelbrich, K. (2014). Identifying appropriate compensation types for service failures: A meta-analytic and experimental analysis. *Journal of Service Research*, 17(2), 195–211.
- Schmitt, B. H., Pan, Y., & Tavassoli, N. T. (1994). Language and consumer memory: The impact of linguistic differences between Chinese and English. *Journal of Consumer Research*, 21(3), 419–431.
- Schwartz, S. H. (2006). A theory of cultural value orientations: Explication and applications. *Comparative Sociology*, 5(2), 137–182.
- Schwartz, S. H. (2011). Studying values: Personal adventure, future directions. *Journal of Cross-Cultural Psychology*, 42(2), 307–319.
- Sheth, J. N. (2011). Impact of emerging markets on marketing: Rethinking existing perspectives and practices. *Journal of Marketing*, 75(4), 166–182.
- Singh, J. (1990). Voice, exit, and negative word-of-mouth behaviors: An investigation across three service categories. *Journal of the Academy of Marketing Science*, 18(1), 1–15.
- Sivakumar, K., Li, M., & Dong, B. (2014). Service quality: The impact of frequency, timing, proximity, and sequence of failures and delights. *Journal of Marketing*, 78(1), 41–58.
- Skitka, L. J., & Tetlock, P. E. (1992). Allocating scarce resources: A contingency model of distributive justice. *Journal of Experimental Social Psychology*, 28(6), 491–522.
- Smith, A. M., & Reynolds, N. L. (2002). Measuring cross-cultural service quality: A framework for assessment. *International Marketing Review*, 19(5), 450–481.
- Smith, A. K., Bolton, R. N., & Wagner, J. (1999). A model of customer satisfaction with service encounters involving failure and recovery. *Journal of Marketing Research*, 36(3), 356–372.
- Tax, S. S., Brown, S. W., & Chandrashekar, M. (1998). Customer evaluations of service complaint experiences: Implications for relationship marketing. *Journal of Marketing*, 62(2), 60–76.
- Triandis, H. C. (1993). Collectivism and individualism as cultural syndromes. *Cross-Cultural Research*, 27(3–4), 155–180.
- Tyler, T. R. (1989). The psychology of procedural justice: A test of the group-value model. *Journal of Personality and Social Psychology*, 57(5), 830–838.
- Vauclair, C.-M., & Fischer, R. (2011). Do cultural values predict individuals’ moral attitudes? A cross-cultural multilevel approach. *European Journal of Social Psychology*, 41(5), 645–657.
- Webster, C., & Sundaram, D. S. (1998). Service consumption criticality in failure recovery. *Journal of Business Research*, 41(2), 153–159.
- Weun, S., Beatty, S. E., & Jones, M. A. (2004). The impact of service failure severity on service recovery evaluations and post-recovery relationships. *Journal of Services Marketing*, 18(2), 133–146.
- Wirtz, J., Xiao, P., Chiang, J., & Malhotra, N. (2014). Contrasting the drivers of switching intent and switching behavior in contractual service settings. *Journal of Retailing*, 90(4), 463–480.
- Zeithaml, V. A., Berry, L. L., & Parasuraman, A. (1993). The nature and determinants of customer expectations of service. *Journal of the Academy of Marketing Science*, 21(1), 1–12.
- Zhu, Z., Sivakumar, K., & Parasuraman, A. (2004). A mathematical model of service failure and recovery strategies. *Decision Sciences*, 35(3), 493–525.